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RESEARCH-ARTICLE

Towards LLMs for Sensor Data: Multi-Task Self-Supervised Learning

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Transformer-based LLMs for Sensor Data

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ABSTRACT

LLMs for vision and NLP domain has been popular by the widespread use of ChatGPT and GPT-4. This paper tackles to build LLMs for sensor domain of one-dimensional signals whose downstream task is activity recognition and emotion detection. We propose a new architecture of Transformer-based self-supervised learner which we name SENvT. This SENvT builds the LLMs for sensor data using 7 pretext objectives in multi-task learning together with contrastive learning. Experimental results show these three. First, we obtained better results for contrastive learning and the masked token task but not for other pretext tasks. Second, the masked token task was better in 60% rather than in 10%. Third, the RGW worked best in accuracy while the masked token task worked best in F1.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; **Activity recognition and understanding**.

KEYWORDS

LLMs, multi-task self-supervised learning, activity recognition

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1 INTRODUCTION

Figure 1 illustrates the overview of the topic in this paper. Recent LLMs such as GPT-4 [20] and ChatGPT [16] have shown the effectiveness of large-scale autoregressive Transformers in multimodal domains [32] including language and vision. As these success attract really large domains of researchers, the recent trend goes towards various other data domains as well. Such generalized model is called foundation models [2]. In terms of the sensor signal domain, this paper tackles how to build such LLMs for sensor data in an unsupervised way. However, some common theme in sensor domain such as domain adaptation in the perspectives of sensor fusions will not be considered in this paper. This is a preliminary study of this quest.

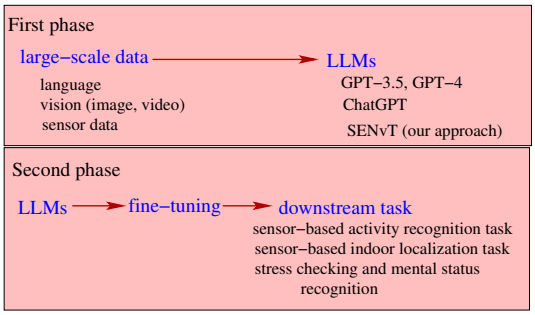

Figure 1: We aim at building LLMs for sensor data. As an extrinsic measure, we recorded the performance of our LLMs on downstream tasks.

Self-supervised learning [8] introduced the idea how to construct large-scale pretrained model without labels and its usage on downstream tasks. For this purpose for vision communities, various kinds of objectives of pretext task[1] are introduced, such as solving the zigsaw puzzle [17], estimating the rotation angle, colorizing an image, inpainting, and so on. Then, vision transformer [9] is introduced as the extension of NLP transformer [27] which is trained with by the objectives of masked token objectives or cloze. DeiT [26] and DINO [4] provide data efficient solution for ViT with distillation and self-supervised learning.

For one dimensional signals, Yuan et al. [30] provides self-supervised learning which investigates the pretext task for time-series data, such as reverse, permute, and time-warp pretext tasks. The idea of these pretext tasks are common to the time-series data augmentation methods [13]. While Iwana and Uchida propose the augmentation methods such as jittering, flipping, scaling, magnitude warping, permutation, window slicing, time warping, and window warping, Yuan et al. used reverse, permute, and time-warp. Zerveas et al. [31] seeks the pretext task of masked token objectives, after preprocessing with sliding window architecture.

*Both authors contributed equally to this research.

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[1]The objectives of pretext task is equally called pretrained objectives. In the case of BERT [5], this objective is equivalent to the Masked Language Models and next sentence prediction tasks.

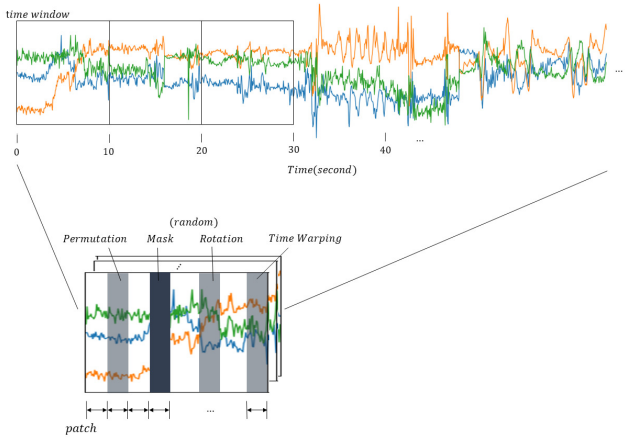

Figure 2: This is an overview figure for multi-task pretext tasks. A sliding window is divided into 10 patches periods where each patch is assigned one of seven pretext tasks randomly. The details of how these patches are assigned are different in patch modes.

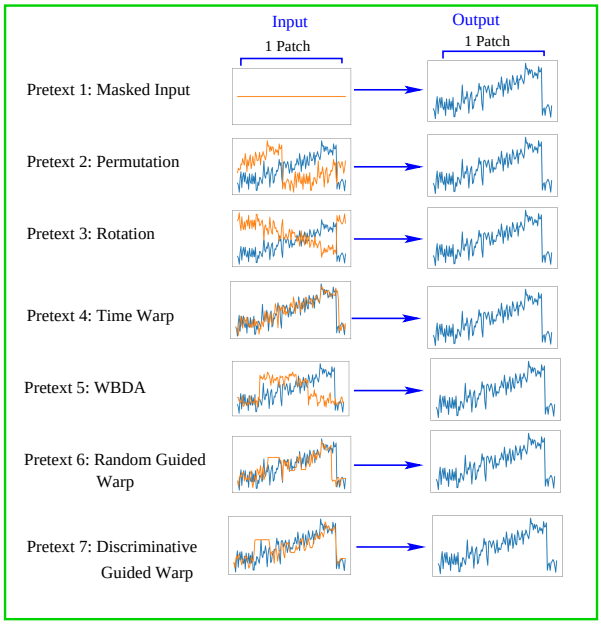

Figure 3: The figure depicts seven modification methods which we used in our pretext tasks. Each pretext task is defined in the following. First, the output is the original signals where the input is modified using the original signals by the masked input, permutation, rotation, time warp, WBDA, random guided warp, and discriminative guided warp. The pretext task is to recover the original signals from the modified signals. Hence, the figure shows 7 pretext tasks where each one recovers the original signal.

It is reported in NLP communities that one of the effect of multi-task pretext settings compared with that of single pretext task is to increase the overall performance, but as well as to let the models robust.

The contribution of this paper is the following:

- We propose SENvT which is a Transformer-based architecture to build LLMs for sensor data in an unsupervised manner. This SENvT learns sensor data with a self-supervised multi-task learning way,
- We propose seven objectives/pretext tasks for SENvT, and
- We measure the performance of SENvT using the performance on downstream task of activity recognition using 5 datasets.

This paper is organized as follows. Section 2 proposes our method. Sections 3 and 4 provide our experimental data and results. Section 5 concludes and provides avenues for further research.

2 OUR METHODS

Self-attention based Transformer models generally operate in a two-stage training procedures. First, pre-training is performed on a large-scale dataset in either a supervised or a self-supervised manner. Using the 1d signals such as tri-axial accelerometer signals, we build the LLM. The objectives of such learning for LLMs are often called a pretext task, which is often operated in without labels. Second, the pre-trained weights are adapted to the downstream tasks using small-mid scale datasets. The downstrem tasks of LLMs for sensor data is sensor-based activity recognition task, for example.

LLMs are built individually before applying to the downstream tasks. Figure 2 shows the overview of our Sensor-based Transformer (we call this SENvT from now on). The SENvT build our sensor-based LLM. The rectangle in the upper figure shows the traditional sliding window architecture for time-series data where this sliding window is fixed-length windows which shows a unit of recognition.

Figure 3 shows seven kinds of pretext tasks. The lower part of Figure 2 shows the sliding window is divided into ten patches where each patch is assigned randomly one of seven pretext tasks or one of five pretext tasks.

2.1 Objectives of Pretext Tasks

We set up the set of objectives of multi-task pretext task are 7 kinds which are illustrated in Figure 3: (1) masked token, (2) permutation, (3) time warping, (4) rotation, (5) wDBA (weighted Dynamic time warping Barycentric Averaging), (6) RGW (Random Guided Warp), and (7) DGW (Discriminative Guided Warp).

Masked token are the original task which was introduced as the cloze task in NLP. Signals are randomly masked token in its input while the target is to recover the signals in the original form. The permutation divides the signal into (four) chunks and randomly shuffles them. The time Warping stretches or shrinks arbitrary segments of a signal. The rotation rotates around the axis of the value zero of the signal.

The last three are pattern mixing methods. In pattern mixing methods, multiple samples of the same class are mixed instead of adding random transformations. The weighted DBA (wDBA) uses weighted version of DTW Barycentric Averaging (DBA) to

generate patterns where DBA is an iterative method of using DTW to align time series to find an average pattern with the features preserved. The last two are related to the guided warping which combines the idea of time warping with pattern mixing. The guided warping warps the features of a sample to the time step relations of a reference, in order to augment the training dataset with new samples. The Random Guided Warp (RGW) selects two intra-class patterns and warps the features of one pattern by the time steps of the second. The Discriminative Guided Warp (DGW) uses a discriminative teacher as the reference for guided warping. The crucial difference from the first four pretext tasks and the latter three tasks (wDBA, RGW, and DGW) are that the data transformations require labels. The latter three require the variance of the signals which can be calculated using labels. The latter three are trained after we preprocess the data. The inputs are the modified signals while the outputs are the original signals.

Modification is made for each patch in the slideing window. The data used this time is 30 Hz and 10 seconds sliding window. Hence, the data format is 300 units and 3 axes. The patch size is set to 30. There are 10 patches per sliding window. The data transformation is applied to 4 randomly selected patches.

The SENvT-u4 and SENvT-m4 use the first four pretext tasks. The SENvT-u7 uses the first seven pretext tasks. It is noted that for the masked token task, the rate of this is 1/10 since we choose one mask among ten slots in the case of task (1). SENvT-u1-X uses the X%. That is, SENvT-u1-10 uses the 10% of masked token.

2.2 Contrastive Learning

Our contrastive learning approach is analogous to the one deployed to DINO, which learns visual features in an online fashion without supervision. The mechanism in DINO is to implement the contrastive learning in the type of BYOL [11] which uses only positive examples for contrastive learning while the SimCLR uses both of positive and negative examples. It uses views of multiple images by comparing their cluster assignments instead of their features. More concretely, via assistance of data augmented versions of the images, it predicts this code from other data augmented versions of the same image. The concrete data augmentation in DINO uses horizontal flip, color jitter, grayscale, and blur.

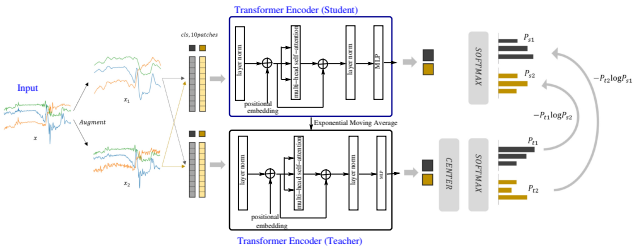

Figure 4: Figure shows the architecture of SENvT which is used to build LLMs for sensor data. It shows our contrastive learning used in our architecture.

Our mechanism is to use the data augmentation for signals such as jitter, scaling, and time warp and do the smoothing of the signals using a Savitzky-Golay filter (savgol filter). The *swapped* prediction problem uses the following loss function: $L(z_t, z_s) = \frac{1}{2}\ell(z_t, q_s) + \frac{1}{2}\ell(z_s, q_t)$ where z_t and z_s comes from two different augmentations of the signals and q_t and q_s are called codes which are the predictions of different image views. For given raw signal data, we parepare two different data augmentations of this signal. We use these augmented signals for the inputs for student and teacher networks of Transformers.

2.3 Transformer

We assume the input data comes in an univariate manner and in a multivariate manner. For this purpose, we design two different kinds of Transformer [27]. For the univariate version, we borrow mechanism from DINO v1 [4] which is a variants of ViT [9]. For the multivariate version, we take mechanisms from MVTS Transformer [31], which can handle the multivariate time series data.

2.3.1 SENvT for Univariate Variables. Since ViT [9] concluded that transformers do not generalize well when trained on inssufficient amounts of data and requires extensive computing resources, DeiT [26] proposes the distillation approach to this. DINO [4] follows the implementation used in DeiT, as well as it adopts the self-supervised approaches. While the ViT architecture takes as input a grid of non-overlapping contiguous image patches of resolution N × N, DINO typically takes N = 16 or 8. The patches are then passed through a linear layer to form a set of embeddings. An extra learnable token to the sequence is added whose rol is is to aggregate information from the entire sequence. The projection head h is attached at its output. Since DINO handles 2 dimensional image dividing into patches, we modifed the internal scan to do the 1 dimensional scan for DINO in the following way. Three parameters in parenthesis denotes the shape of the data which is entered appropriately.

Algorithm 1 Algorithm shows a pseudo algorithm for SENvT for univariate variables

-PatchEmbed embeds the data [batch=1500, channel=3, length=300] after patch-by-patch deformation in each operation with one-dimensional convolution [1500, patch=10, embed_dim=768].
-Add cls_token representing window information [1500, 11, 768].
-Add position information with positional encoding using trigonometric functions [1500, 11, 768].
-Then it passes through 12 Block layers with 12-head MultiHeadAttention [1500, 11, 768],
-Pass the information [1500, 1, 768] of cls_token to the final layer.
-The final layer has two linear layers with a ramp function between them.
-Set the output in the second linear layer to 3, which represents the number of channels, so that the output is of the same form as the input data [1500, 3, 300].
-300 sets of this are created and output in the form of [1500, 3, 300]. Measure and update the MSELoss between this output and the original signal data before data transformation.

2.3.2 SENvT for Multivariate Variables. First, this transformer treats the multivariate time series data. Suppose that the $X \in \mathbf{R}^{w \times m}$ is a multivariate time series of length w. This sample contains m

Training Data for Pretext Task					
Dataset	Subjects	Samples	Classes	Environment	References
Capture-24	152	573K	4	Free-living	Willetts et al. (2018) [29]
Test Data for Downstream Task					
Dataset	Subjects	Samples	Classes	Environment	References
ADL	7	0.6K	5	Lab	Bruno et al. (2013)[3]
Opportunity	4	3.9K	4	Semi free-living	Roggen et al. (2010)[22]
PAMAP2	8	2.9K	8	Lab	Reiss and Stricker (2012)[21]
REALWORLD	14	12K	8	Lab	Sztyler and Stuckenschmidt (2016)[24]
WISDM	46	28K	18	Semi free-living	Weiss et al. (2019)[28]

Table 1: Dataset used for pretext and downstream tasks

different variables. We see each sample at time i, containing m different variables, as one feature. The feature fectors x_t are first normalized (for each dimension, we subtract the mean and divide by the variance across the training set samples) and then linearly projected onto a d-dimensional vector space, where d is the dimension of the transformer model sequence element representations (model dimension) by $u_t = W_p x_t + b_p$ where $W_p \in \mathbf{R}^{d \times m}$, $b_p \in \mathbf{R}^d$. W_p, b_p, and u_t corresponds to queries, keys and values of the self-attention layer. For the positional encodings, mvts uses the fully learnable positional encodings while the original uses the deterministic sinusoidal encodings. Positional encodings are learned so as to occupy a different, approximately orthogonal, subspace to the one in which the projected time series samples reside.

3 EXPERIMENTAL DATA

Table 1 shows the summary of dataset used for pretext and downstream tasks. For pretext tasks for self-supervised learning, capture-24 was used after transforming into unlabeled data[2]. When we used this dataset for three pretext tasks which require labels, we used the same dataset since this data has labels. The accelerometer in capture-24 dataset [10] was attached on a lanyard or clipped to his/her wrist. It contained 124 participants with more than 2,500 hours, which was collected at Oxford surroundings for daily living. One strong belief that the organizers of this collection is that one accelerometer is sufficient settings for doing activity recognition while many others tried to collect other sensor data such as gyroscope, magnatometer and so on.

For downstream tasks, we used the 5 kinds of dataset including ADL [3], Opportunity dataset [22] PAMAP2 [21], RealWorld [24], and WISDM [28]. We did not use data augmentation for downstream tasks. Table 1 shows a summary of data.

The test data and training data were divided into 0.2 and 0.8, and the training data was further divided into validation data 0.2 and training data 0.8. Since the class variation was conspicuous in the data used, the weight of the class was calculated and the loss was calculated considering the variation of the class.

[2]ssl-capture-24:https://doi.org/10.5281/zenodo.6574265

4 EXPERIMENTAL RESULTS

Table 3 shows the preliminary results of extrinsic evaluation of the quality of LLMs. This is since the intrinsic evaluation measure is not obtained or not established for this domain[3].

In Table 3, SSL uses the ResNet for the self-supervised learning [30]. Hence, this is not Transformer-based architecture. SENvT-u and SENvT-m are based on our proposal in this paper (until up to now we call these simply SENvT) where SENvT-u handles univariate settings and SENvT-m handles multivariate settings. SENvT-u4 (and SENvT-m4) stands for the 4 pretext settings: masked token, permutation, time warp, and rotation tasks. SENvT-u7 stands for the 7 pretext settings: masked token, permutation, time warp, rotation, wdba, rgw, and dgw tasks. Our pretext tasks were arranged in the way that (1) the sliding window was of the size 300 (2) 10 patches were taken from this sliding window, and (3) 10 patches were randomly selected by 4/7 pretext tasks.

We have two kinds of results: fine tuning and transfer learning. Transfer learning refers to the method where the last layer is differentiable. Fine tuning refers to the method where all the layers are differentiable. We set the learning rate for transfer learning 0.0001 and that for fine tuning 0.00005.

SENvT-u uses the AdamW optimizer while SENvT-m uses the rectifid Adam (RAdam) optimizer. Other settings are, epoch = 100, learning rate = 0.0001, loss function = cross entropy, class weight = true, and batch size = 1500. The best epoch in each learning was selected as the one with the highest percentage of correct answers in the validation data. Using the model, accuracy and f1 scores for the test data are summarized in the form of (mean ± std).

4.1 Comparison among Pretext Tasks

Table 4 shows the comparison of the effect of pretext tasks as a standalone pretext task. Only each pretext task is applied measureing the performance on the downstream task, i.e. extrainsic task. We used the opportunity dataset for this purpose. If we consider the accuracy as the measure, the order of effective pretext task was RGW (1st), masked token (2nd), permutation (3rd), rotation (4th), DGW (5th), wBDA (6th), and time warp (7th). In we consider F1, the order was masked token (1st), permutation (2nd), DGW (3rd), time warp (4th), rotation (5th), RGW (6th), and wBDA (7th).

[3]For the language domain, the perplexity measure is established.

	fine-tune		transfer-learn	
	acc	f1	acc	f1
SSL (ResNet)[30]				
ADL	.965±.016	.955±.023	.841±.006	.805±.012
Opportunity	.842±.005	.855±.009	.671±.016	.629±.006
PAMAP2	.908±.008	.907±.008	.793±.007	.796±.006
RealWorld	.960±.001	.965±.001	.837±.004	.850±.004
WISDM	.937±.003	.937±.003	.744±.002	.742±.003
SENvT-u4 (Transformer)				
ADL	.909±.012	.878±.015	.839±.015	.794±.012
Opportunity	.749±.019	.741±.021	.710±.007	.683±.017
PAMAP2	.849±.010	.845±.010	.716±.003	.691±.004
RealWorld	.902±.005	.912±.005	.747±.005	.751±.005
WISDM	.869±.005	.868±.004	.667±.002	.662±.003
SENvT-u7 (Transformer)				
ADL	.899±.010	.868±.012	.847±.011	.785±.022
Opportunity	.743±.009	.745±.015	.715±.007	.691±.011
PAMAP2	.848±.012	.845±.014	.710±.005	.687±.005
RealWorld	.906±.003	.916±.003	.763±.003	.769±.003
WISDM	.888±.002	.887±.002	.682±.003	.677±.004
SENvT-contrastive (Transformer)				
ADL	.902±.019	.878±.021	.884±.023	.869±.021
Opportunity	.766±.003	.774±.007	.756±.008	.768±.007
PAMAP2	.834±.013	.822±.013	.811±.006	.798±.007
RealWorld	.888±.004	.898±.004	.812±.002	.824±.002
WISDOM	.869±.002	.868±.002	.775±.002	.772±.002
SENvT-m4 (Transformer)				
ADL	.921	.917	.890	.890
Opportunity	.699	.682	.672	.626
PAMAP2	.704	.696	.676	.658
RealWorld	.806	.806	.733	.729
WISDOM	.766	.767	. 642	.637
SENvT-m4 (Transformer)$^{4-patch-version}$				
ADL	.866	.847	.850	.850
Opportunity	.698	.655	.687	.676
PAMAP2	.700	.691	.657	.646
RealWorld	.800	.799	.726	.718
WISDOM	.750	.753	.640	.637

Table 2: Table shows the results by our SENvT-u4/SENvT-u7/SENvT-m4 compared with SSL (ResNet) [30]. Transfer learning approach for our SENvT-u7 and SENvT-contrastive outperform SSL (ResNet). On the other hand, fine-tuning, results of SSL (ResNet) are better than those of Transformer.

4.2 Overall Performance

Table 3 shows the overall performance of SENvT Transformer compared with ResNet. In terms of transfer learning, 8 out of 10 items in SENvT-contrastive were better than ResNet. While in terms of fine tuning, all the items in ResNet were better than SENvT-X.

SENvT-u4 and SENvT-u7 were less powerful compared with SENvT-contrastive and SENvT-u1-X. This suggests us that more difficult task such as masked token and contrastive learning got better results than semantically meaningful pretext tasks such as permutation, rotation, time warp, and so on.

SENvT-u-p1-X (Transformer)				
Opportunity	acc	f1	acc	f1
SENvT-u1-p1-10	.724	.735	.633	.636
SENvT-u1-p1-40	.746	.732	.693	.669
SENvT-u1-p1-50	.747	.743	.687	.635
SENvT-u1-p1-60	.765	.773	.662	.631
SENvT-u1-p1-70	.704	.667	.685	.623

Table 3: Comparison among SENvT-u1-p1-X (pretext task 1) where the percentage of masked token is variable. 60% is best in fine tuning while 40% is best in transfer learning.

SENvT-u1-pX (Transformer)		
fine-tune		
Opportunity	acc	f1
SENvT-u1-p1	.676	.657
SENvT-u1-p2	.671	.621
SENvT-u1-p3	.664	.633
SENvT-u1-p4	.646	.623
SENvT-u1-p5	.656	.602
SENvT-u1-p6	.680	.616
SENvT-u1-p7	.658	.627

Table 4: Comparison among pretext tasks 1 to 7. The performance is measured using opportunity dataset. Pretext task 6 works best in accuracy while pretext 1 works best in F1.

SENvT-m4 is less powerful except ADL dataset, compared with SENvT-u4 with same set of pretext tasks. Although this architecture made connections among the variables in the same time tick, we had only three variables. This might be the reason for this result. If we had more variables, results might be different.

The comparison among the various percentage of masked token pretext task was obtained in SENvT-u1-X. As MAES [12] showed that the percentage of masked token (in our case, this corresponds to Pretext 1) of 75% is better than that of 15% in BERT [5]. For the comparison of the rate of pretext task 1, SENvT-u1-10 stands for 10% of the rate of masked token.[4] In fine-tune, SENvT-u1-60 (60%) is best among others. In transfer learning, SENvT-u1-40 (40%) is best among others.

In fine tuning, SENvT-u4 and SENvT-u7 were less powerful compared with ResNet. One possible reason was that the number of parameters in Transformer are much bigger than ResNet. Our experiments were based on capture24 dataset of 573k data points. This can be large scale as a sensor-based dataset but not a huge/gigantic dataset over a million or a trillion data. Similar fix was reported in the case of image/video in vision community [15]. If we do the similar strategy, results might be different.

5 RELATED WORKS

Although self-supervised learning appeared in various domains for language and vision, a very few papers have been appeared

[4]Similarly, SENvT-u1-X stands for X% of the rate of masked token.

for one dimensional signal domain. A very few approaches of self-supervised learning include [6, 30] where the downstream task of [30] is sensor-based activity recognition and that of [6] is emotion recognition. The approach of [30] uses ResNet while that of [6] uses inception-inspired neural networks. They do not use Transformer like us. The pretext task of [30] uses the smaller set, i.e. pretext tasks 2, 3, and 4 in SENvT. That of [6] uses different kinds of pretext tasks, which are mainly classification-oriented pretext tasks.

6 CONCLUSIONS

This paper aims at building LLMs for sensor data, focusing on how to build such LLMs. We propose a new architecture of self-supervised learner SENvT in an unsupervised manner with multi-task self-supervised learning, proposing 7 pretext objectives and contrastive learning. First, we obtained better results for contrastive learning and for the masked token pretext task but not for other pretext tasks. Second, the masked token pretext task is better in 60% rather than in 10%. Third, among pretext tasks, the RGW works best in accuracy but the masked token in F1.

We have two further avenues. First, this paper omits the difficulty to use multiple sensor datasets especially which attached in different locations as the training data. This will require the study how to resolve the sensor fusion: varieties of sensor types (accelerometer, gyroscope, MOCAP, and video) with multiple subjects. Second, we obtained better results for transfer learning but worse results for fine tuning compared with ResNet. As the lesson from the vision community where the similar situations arised, it was pointed out that one crucial defect of Transformer was the extremely bigger parameter space compared with ResNet [15]. We might be solve this problem in a similar manner.

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